MATH 11 — ELEMENTARY STATISTICS

§6.3

The Standard Normal Distribution

Normal distributions, probabilities and critical values

Today's Goal

After completing this section, you should be able to:

- Understand the properties of normal distributions
- Use the standard normal distribution to find probabilities
- Use Table A-1 and Excel to find probabilities from z-scores
- Find z-scores from known probabilities
- Identify and calculate critical values

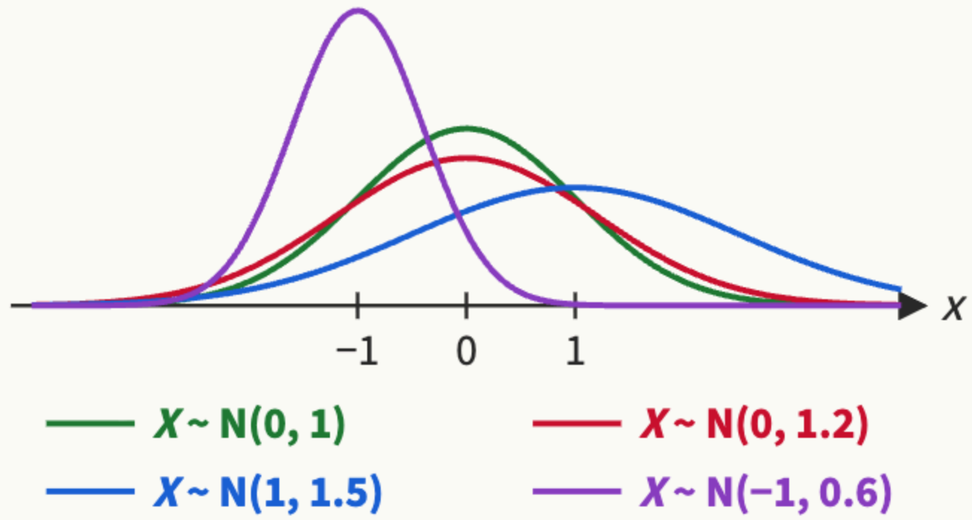
Learning Target 10

LT.10 Normal Probability Distributions

I understand what a normal probability distribution is. I can distinguish the difference between various normal probability distributions. I can use the standard normal distribution appropriately to calculate the probability of an event and to determine appropriate critical values.

Normal Distributions

The notation $X \sim N(\mu, \sigma)$ means that X is a random variable for a (continuous) normal distribution where μ and σ are the mean and standard deviation of X.



What does μ change? What does σ change?

Notice that μ is where the distribution is centered and the value of σ determines how spread out the distribution is.

Normal Distribution

Definition 6.3.1

A **normal distribution** is a probability distribution with a continuous random variable whose graph is symmetric and bell-shaped. The probability density function of a normal distribution is

$$f(x) = \frac{e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}}{\sigma\sqrt{2\pi}}$$

(You do not need to memorize this function.)

Standard Normal Distribution

Definition 6.3.3

The **standard normal distribution** is a normal distribution where $\mu = 0$ and $\sigma = 1$. So $X \sim N(0, 1)$.

If $X \sim N(\mu, \sigma)$, then the z-score associated with a value of X is given by

$$z = \frac{x - \mu}{\sigma}.$$

When $X \sim N(0, 1)$, then we get $z = x$. In other words, z-scores can be thought of as a continuous random variable that is normally distributed with a mean of 0 and standard deviation of 1.

Standard Normal Probabilities

Procedure 6.3.4 Using Excel for Standard Normal Probabilities

The Excel function `NORM.DIST` calculates probabilities for normal distributions:

```
=NORM.DIST(x, mean, standard_dev, TRUE)
```

where

- `x` = the value of x (or z-score for the standard normal distribution);
- `mean` = the mean of the distribution (use 0 for the standard normal distribution);
- `standard_dev` = the standard deviation (use 1 for the standard normal distribution);

Standard Normal Probabilities

Procedure 6.3.4, continued

- `TRUE` = cumulative probability (always use `TRUE`).

Important: this function calculates $P(X \leq x)$, the probability to the *left* of x.

Examples:

- $P(z < -0.56)$: `=NORM.DIST(-0.56,0,1,TRUE)`
- $P(z > 0.34)$: `=1-NORM.DIST(0.34,0,1,TRUE)`

Find Each Probability

Since normal distributions are examples of continuous probability distributions, there is still a correspondence between area and probability.

To determine probabilities for regions defined by z-scores, we use either Table A-1 (the Standard Normal Distribution Table) or technology (Excel or an online calculator).

Find each of the following probabilities.

(a) $P(z > 0)$

(b) $P(z < -1.38)$

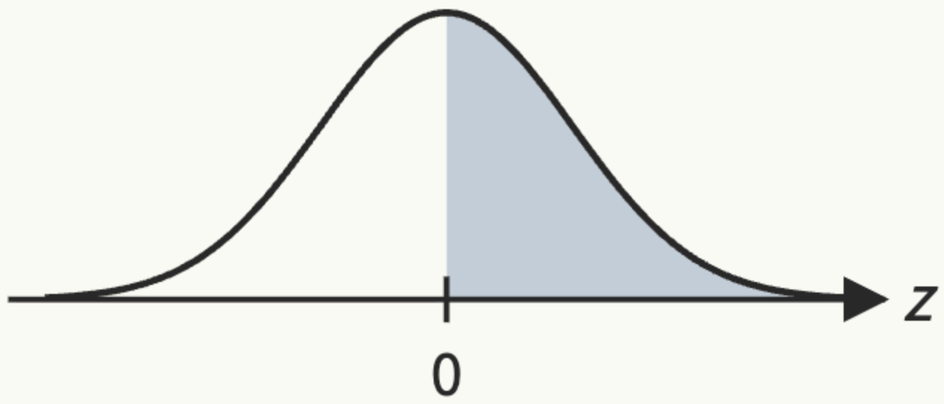
(c) $P(z > 0.71)$

(d) $P(-1.25 < z < 0.62)$

Part (a): Half the Area

(a) $P(z > 0)$

Since the standard normal distribution is symmetric and centered at 0, the region where $z > 0$ coincides with exactly half of the area under the curve.



So $P(z > 0) = 0.5$.

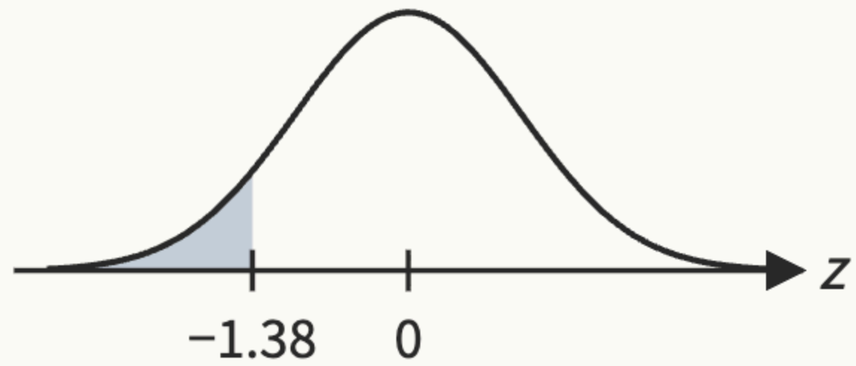
Part (b): A Left Tail

(b) $P(z < -1.38)$

Using Table A-1, we start on the page with the negative z-scores. Then we go down to the row labeled "-1.3" and then move across to the column labeled "0.08". The table entry is 0.0838.

Using Excel, we would enter `=NORM.DIST(-1.38,0,1,TRUE)` into any empty cell.

Either way, $P(z < -1.38) \approx 0.0838$.

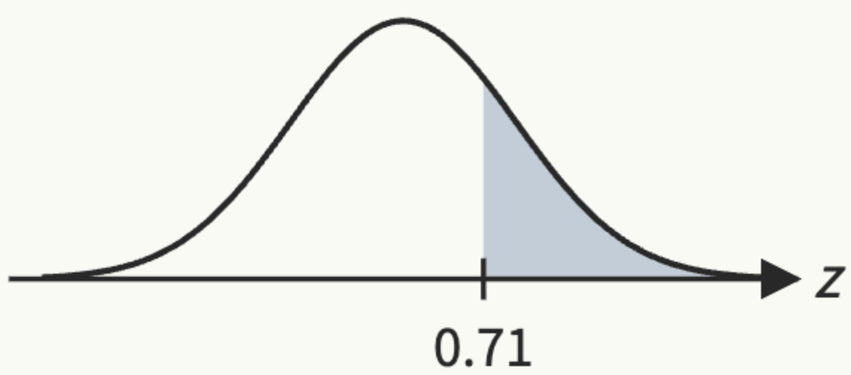


Part (c): A Right Tail

(c) $P(z > 0.71)$

Table A-1 always reports area to the *left*. There are two ways to handle a right-tail question:

1. By symmetry, $P(z > 0.71) = P(z < -0.71)$. Look up -0.71 in the negative table.
2. Use the complement: $P(z > 0.71) = 1 - P(z < 0.71)$. From the positive table, $P(z < 0.71) = 0.7611$, so $P(z > 0.71) = 1 - 0.7611 = 0.2389$.



Part (c): Using Excel

(c) $P(z > 0.71)$

Using Excel, either:

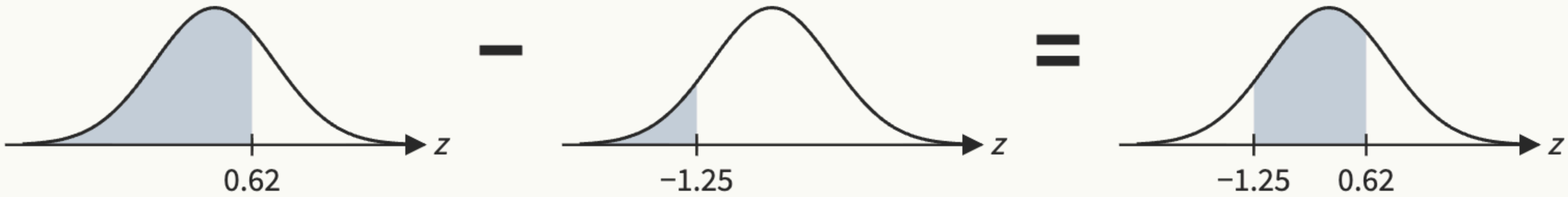
- Method 1: `=NORM.DIST(-0.71,0,1,TRUE)`;
- Method 2: `=1-NORM.DIST(0.71,0,1,TRUE)`.

So $P(z > 0.71) \approx 0.2389$.

Part (d): Between Two Values

(d) $P(-1.25 < z < 0.62)$

This is the probability that z is between -1.25 and 0.62. Use Table A-1 by subtracting two left-tail areas:



0.7324 **0.1056** **0.6268**

So $P(-1.25 < z < 0.62) = 0.7324 - 0.1056 = 0.6268.$

Part (d): Using Excel

(d) $P(-1.25 < z < 0.62)$

Using Excel, enter `=NORM.DIST(0.62,0,1,TRUE)-NORM.DIST(-1.25,0,1,TRUE)`. Excel returns the more precise value 0.62672133, which rounds to 0.6267, differing slightly from the table because table values are rounded.

Using Excel to Find z-scores

Procedure 6.3.6 Using Excel to Find z-scores

The Excel function `NORM.INV` finds the z-score (or x-value) for a given cumulative probability:

```
=NORM.INV(probability, mean, standard_dev)
```

where

- `probability` = the cumulative probability (area to the left);
- `mean` = the mean of the distribution (use 0 for the standard normal distribution);
- `standard_dev` = the standard deviation (use 1 for the standard normal distribution).

Using Excel to Find z-scores

Procedure 6.3.6, continued

Example: to find a where $P(z < a) = 0.8907$, enter `=NORM.INV(0.8907,0,1)`.

Find the Value of a

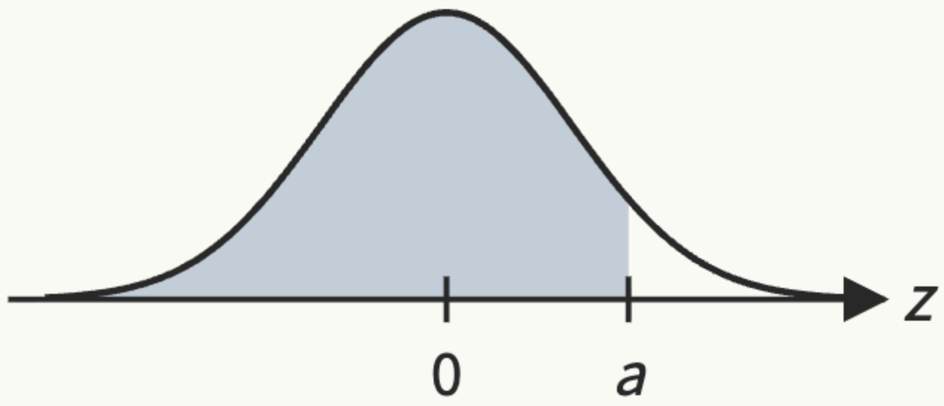
(a) Find the value of a such that $P(z < a) = 0.9265$.

Since the area to the left is greater than 0.5,

$a > 0$.

Look up the area 0.9265 in Table A-1: it sits at row "1.4", column "0.05". So $a = 1.45$.

Excel: `=NORM.INV(0.9265,0,1)`.



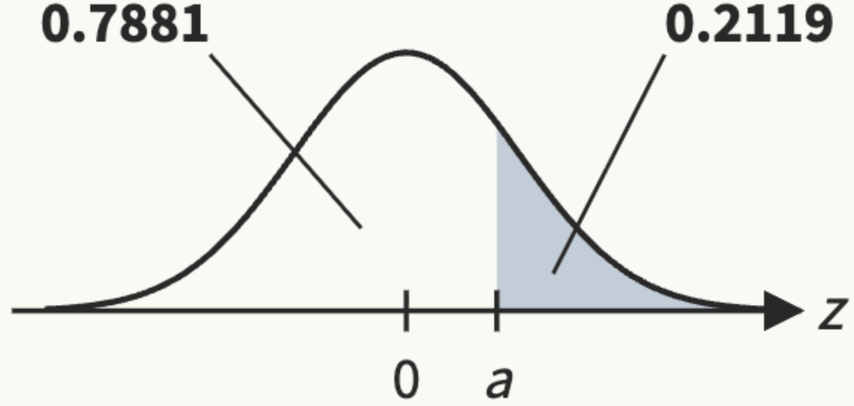
Part (b): Area to the Right

(b) Find the value of a such that $P(z > a) = 0.2119$.

Since the area of the region extending to the right is 0.2119, which is less than 0.5, we know that $a > 0$. Also, we know that the area of the region extending to the left must be $1 - 0.2119 = 0.7881$.

Using Table A-1, we look for the area of 0.7881, which corresponds to $z = 0.80$. So $a = 0.80$.

Excel: `=NORM.INV(0.7881,0,1)`.



Critical Value

Definition 6.3.8

For the standard normal distribution, a **critical value** is a z-score on the borderline separating those z-scores that are "significantly low" or "significantly high".

We use the notation z_α to denote the z-score with an area of α to the right. So $P(Z > z_\alpha) = \alpha$ and equivalently $P(Z \leq z_\alpha) = 1 - \alpha$.

Find a Critical Value

Find the value of $z_{0.10}$.

The value we are looking for is the z-score that has an area of $\alpha = 0.10$ to the right. This means that the area to the left is $1 - \alpha = 0.90$.

Using Table A-1, we look for the area of 0.90. The closest value we can find is 0.8997, which corresponds to $z = 1.28$. So $z_{0.10} \approx 1.28$.

Excel: `=NORM.INV(0.90,0,1)` returns 1.281551566, which rounds to 1.28.

